BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

MEDICAL DISCLAIMER: AI-based preliminary estimation only. This result must be confirmed using a standard laboratory blood typing test before medical use. This system does not replace standard serological blood typing.

Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%)

1. EXAMINATION & SAMPLE AUDIT DETAILS

Report ID:	BG-000400	**Timestamp:**	2026-09-06 19:21:21
Input Source:	Mantra MFS100 Scanner | Standard Finger: LEFT THUMB	**Evaluation Status:**	LOW_CONFIDENCE
Model Name:	Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%)	**Confidence:**	**39.68%**
Standard Acquisition Finger:	**LEFT THUMB** (Convention-based protocol — not automatically verified by scanner)		

2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP

Predicted Blood Group:

AB+

Confidence Score: **39.68%**

Threshold: 60.0%

Classification: Low Confidence - Retest Required

Derived ABO Group: **AB** (48.5% prob) | Rh Factor: **+**

Rh FACTOR UNCERTAINTY ADVISORY: Rh factor (±) determination is uncertain (moderate 8-class confidence (39.7%)). ABO group appears to be 'AB' (48.5% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

A+	A-	AB+	AB-	B+	B-	O+	O-
4.7%	1.4%	39.7%	8.8%	22.0%	2.9%	1.9%	18.7%